EVERY ALGEBRAIC STACK CONTAINS LARGEST OPEN SUBSTACKS THAT ARE DM AND QUASI-DM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The characterization of the DM locus through the existence of a surjective etale cover by a scheme. Invoked inside the counted proof (source lines 4807-4810). Supplies the presentations that make the union of two open DM substacks again DM.
- The same characterization for the quasi-DM locus. Invoked at the end of the counted proof (source lines 4812-4813). Repeats the argument verbatim in the quasi-DM case.

Source reference. Original theorem theorem-DM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 610–610. Navigation pointer only: the counted text refers to the Stacks item theorem-DM, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original theorem theorem-quasi-DM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 619–619. Navigation pointer only: the counted text refers to the Stacks item theorem-quasi-DM, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. There exist open substacks*

$$\mathcal{X}'' \subset \mathcal{X}' \subset \mathcal{X}$$

such that $\mathcal{X}''$ is DM, $\mathcal{X}'$ is quasi-DM, and such that these are the largest open substacks with these properties.

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Proof. All we are really saying here is that if $\mathcal{U} \subset \mathcal{X}$ and $\mathcal{V} \subset \mathcal{X}$ are open substacks which are DM, then the open substack $\mathcal{W} \subset \mathcal{X}$ with $|\mathcal{W}| = |\mathcal{U}| \cup |\mathcal{V}|$ is DM as well. (Similarly for quasi-DM.) Although this is a cheat, let us use Theorem theorem-DM to prove this. By that theorem we can choose schemes U and V and surjective étale morphisms $U \to \mathcal{U}$ and $V \to \mathcal{V}$. Then of course $U \amalg V \to \mathcal{W}$ is surjective and étale. The quasi-DM case is proven by exactly the same method using Theorem theorem-quasi-DM. $\square$